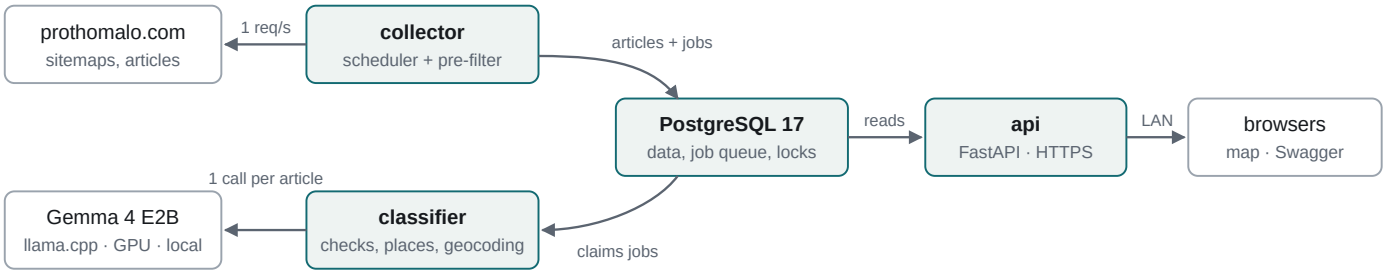
Dhaka Crime Map — Pipeline Report

26 October 2026 · in-house deployment on **genweb2inhouse** (192.168.1.200) · repository github.com/zawadgenweb2/dhaka-crime-map · status: **live**

1. What it does

- Collects crime news from **Prothom Alo** (Bangla) twice a day (08:00 and 20:00 Asia/Dhaka), plus on-demand backfills of past days.
- An **in-house language model (Gemma 4 E2B)** reads each likely crime article and extracts incidents: crime type, place in Dhaka district, date, time of day, victims and a verbatim supporting quote.
- Incidents that pass automatic checks appear on a bilingual (Bangla/English) **live heat map** with filters, hotspots and a news feed; every point links to its source article. No external AI service is used.

2. Architecture



Five Docker services (db, migrate, api, collector, classifier) plus one systemd model server, all on one in-house GPU server (RTX 3080 Ti, 12 GB). Postgres is the database, job queue and lock service; the database exists only on this server.

3. Pipeline, step by step

#	STEP	HOW	RESULT
1	Discover	Prothom Alo news sitemap (last 48 h, with titles) + one daily sitemap per day; robots.txt honoured; 1 request/s	~300 articles/day listed
2	Pre-filter, before download	Crime & capital sections always; any other section if the **title or keywords contain a crime word**; videos, galleries, sister sites never	~10% downloaded
3	Pre-filter, full text	Needs a crime word and a **Dhaka-district place** in the text (namesakes like "কুষ্টিয়ার মিরপুর" and "sent to Dhaka Medical" do not count)	~20–30 articles/day to Gemma
4	Classify	One Gemma call per article: full text + the Dhaka places it mentions + an 8-day Bangla calendar; answer forced into a JSON schema	incidents, ~3 s/article
5	Check & locate	Quote must be verbatim in the article; place must be named where the crime is described; first sentence must not put it in another district; no future dates	published or held
6	Group duplicates	Same category + place, dates ±1 day, compatible time of day	one point per event
7	Exact spot	Street/landmark geocoded via OpenStreetMap (1 request/15 s), kept only within 2.5 km of the place	refined points
8	Serve	REST API + dashboard; the page updates itself when data changes; per-day coverage shows how complete a period is	live map

4. Model and quality

Gemma 4 E2B (Q8 GGUF) on **llama.cpp**, CUDA, loopback only with an API key, always loaded (2.9 GB GPU memory). Measured on 71 reference articles labelled in the prototype phase:

MEASURE	RESULT
Valid structured answers	100%
Article: crime in Dhaka? precision / recall	0.87–0.90 / 0.81
Incidents: precision / recall	0.80–0.83 / 0.64–0.66
Thana / place / category / date correct	0.93 / 0.86 / 0.76 / 0.55–0.63
Time per article (median)	2.4 s

Engine/model benchmark (same articles and prompt):

SETUP	INCIDENT P / R	DECODE	GPU MEM
E2B, llama.cpp (chosen)	0.80–0.83 / 0.65	120 tok/s	2.9 GB
E4B (QAT Q4), llama.cpp	0.71 / 0.69	113 tok/s	3.5 GB
E4B (QAT w4a16), vLLM	stopped	~120 tok/s	11.7 of 12

vLLM gave no speed gain for one-at-a-time requests and needed nearly the whole card; the larger E4B invented more incidents. Prefill ≈ 4,400 tok/s; first token ≈ 0.14 s.

5. Current status (26 Oct, 15:25 Dhaka)

213	169	360	14 MB	75
incidents on the map	in the last 30 days	articles read by Gemma	database size	automated tests passing

The **30-day backfill is running** (about 1,900 more articles to download at 1 per second; Gemma keeps pace). Last 30 days so far: assault 34, murder 30, drugs/weapons 30, theft 26, fraud/extortion 25, kidnapping 13. Most incidents: Shah Ali 13, Mirpur 10, Adabor 10, Cantonment 9, Mohammadpur 8. All earlier prototype (Claude) output was removed from the map; it is kept only as a frozen test set.

6. Safeguards

- **Only in-house Gemma output is served**; every endpoint checks provenance.
- **Evidence:** each incident carries a quote that must appear word for word in the article.
- **Location grounding:** the place must be named where the crime is described; another district in the lead blocks it.
- **Reliability:** every model call is recorded before it is sent and every answer saved, so re-checking needs no new model calls; an older job can never overwrite newer output.
- **Coverage honesty:** each day is marked complete, processing, partial, failed or not collected; the map warns when a period is incomplete.
- **Politeness:** robots.txt, 1 request/s to the publisher, OpenStreetMap ≤ 4/min.
- **Security:** LAN-only HTTPS; admin token for actions that change data or use the GPU; secrets never in git.
- **Backups:** nightly database dump, 14 days kept.

7. Interfaces

READ (OPEN ON THE LAN)	ADMIN (TOKEN)
`/api/incidents`, `/api/stats` (filters: days or dates, crime type, thana) · `/api/coverage` per-day completeness · `/api/meta` status and progress · `/api/runs` history	`POST /api/scrape` collect now (up to 90 days) · `POST /api/classify` try any pasted article, nothing saved · review: fix category/place, reject, retry, dismiss an article

Dashboard https://192.168.1.200:8100 · Swagger https://192.168.1.200:8100/docs · API guide with real responses: docs/API.md · engineering handbook: docs/ENGINEERING.md.

8. Known limits

- One newspaper; these are **reported** incidents, not all crime. Results are published without human review, so some errors reach the map (incident precision ≈ 0.8).
- The model still sometimes splits one event into several incidents (e.g. extortion + kidnapping + assault for one case); dates are the weakest field.
- Older days: Prothom Alo's daily lists have no titles, so past sports/business/world articles are not examined (a 48-hour audit found no Dhaka crime among them).
- Points without a named street sit at the neighbourhood or thana centre. Browser shows "Not secure" (self-signed LAN certificate) until Tailscale HTTPS is enabled.

9. Next steps

ITEM	WHY	EFFORT
Prompt and rule work against over-splitting and dates	largest remaining accuracy gap	days
Second newspaper (Samakal) with per-source provenance	more coverage, cross-checking	1–2 weeks
Any district on demand (national gazetteer ready)	user request	2–3 weeks
Parallel model slots (3–4 at once)	~2× throughput for large backfills	hours
Tailscale HTTPS, long admin token, then public release review	remove browser warning; go public safely	hours + admin